

Olist Marketplace — Drivers of Low Review Scores

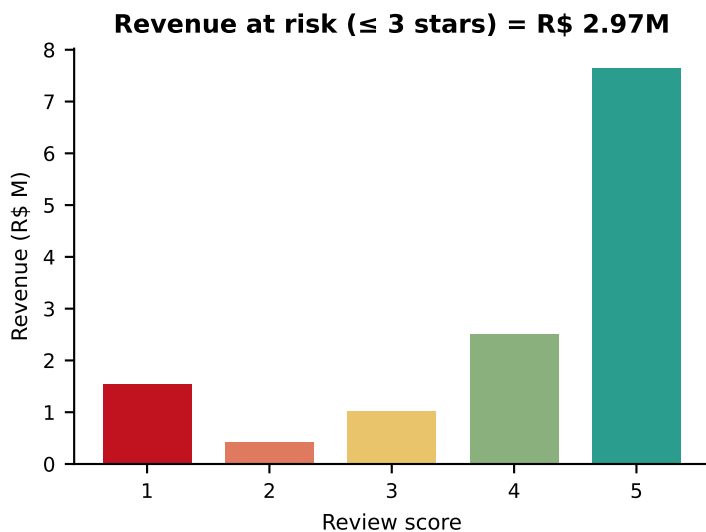
Executive Summary · End-to-end data analysis (EDA → inferential tests → predictive & prescriptive model)

PROBLEM

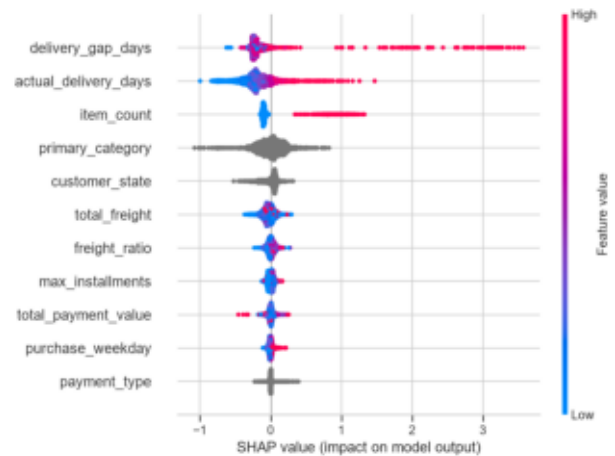
About one order in five (≤ 3 stars, 21% of 95,824 reviewed orders) is low-rated. These orders carry R\$ 2.97M — 22% of item revenue — in revenue at risk, threatening repeat purchase and seller retention.

APPROACH

Cleaned 96k delivered orders; sized the problem in EDA; confirmed drivers with five hypothesis tests; built a Logistic→LightGBM risk model (time-based split) explained with SHAP; simulated a targeted intervention.



Model drivers (SHAP): delivery timing leads



KEY FINDINGS

1. Delivery lateness is the #1 driver.

Low-review orders lose ~5.8 days of early-delivery cushion (Welch $t = 56$, $p \approx 0$); delivery gap & actual days are the top-2 SHAP features.

2. Risk concentrates in a few big categories.

Category means differ significantly (ANOVA $F = 44$, $p \approx 0$). Top revenue-at-risk: bed_bath_table (R\$ 277k), watches_gifts (R\$ 244k), health_beauty (R\$ 229k).

3. Worst states fail on consistency, not just speed.

Delivery-gap variance in the 5 worst states is 2.4× the 5 best (Levene $W = 2527$, $p \approx 0$).

4. A risk score targets intervention efficiently.

LightGBM ROC-AUC 0.66, PR-AUC 0.36; flagging the top-decile gives 2.6× precision (44% vs 17%) and catches 26% of low reviews.

RECOMMENDATIONS

- Target the late tail: alert ops when an order slips past its estimate; fix worst routes/carriers rather than an already-early median.
- Prioritise category quality/logistics on bed_bath_table and computers_accessories (high volume × worst review mix).
- In the worst states, cut delivery variance (carrier SLAs, buffers) — reliability over raw speed.
- Deploy the at-delivery risk score for proactive outreach on top-decile orders (pre-review window).

Deprioritise (low leverage):

freight ratio (Spearman $\rho = -0.03$) and installment count are significant but too small to move the average.

R\$ 2.97M revenue at risk · 21% low-review orders · ROC-AUC 0.66 · 2.6× targeting lift

Source: Olist Brazilian E-Commerce Public Dataset (Kaggle)